

Slope-Intercept and Point-Slope Form

Directions: Show all work. No calculators.

1. Identify the slope and y-intercept of $y = 4x - 7$.
2. Identify the slope and y-intercept of $y = -\frac{3}{5}x + 2$.
3. Write an equation in slope-intercept form with slope -2 and y-intercept 5 .
4. Write an equation in slope-intercept form with slope $\frac{3}{4}$ and y-intercept -6 .
5. Write the equation of the line through $(0, 3)$ and $(4, -5)$ in slope-intercept form.
6. Write the equation of the line through $(0, -4)$ and $(-6, 5)$ in slope-intercept form.

7. Write an equation of the line through $(-3, 4)$ and $(5, 8)$ in point-slope form.

8. Write an equation of the line through $(2, -5)$ and $(6, 7)$ in point-slope form.

9. Write a point-slope equation for the line with slope -4 through $(-2, 7)$.

10. Write a point-slope equation for the line with slope $\frac{5}{3}$ through $(3, -1)$.

11. A line goes through the points $(-6, -5)$, $(-2, 1)$, $(2, 7)$, $(6, 13)$. Write four different point-slope equations for the line.

12. A line goes through the points $(-7, 10)$, $(-4, 4)$, $(-1, -2)$, $(2, -8)$. Write four different point-slope equations for the line.
13. Convert $y - 5 = 3(x + 2)$ to slope-intercept form.
14. Convert $y + 4 = -2(x - 3)$ to slope-intercept form.
15. Convert $y - 1 = \frac{3}{4}(x + 8)$ to slope-intercept form.
16. Convert $y + 6 = -\frac{2}{3}(x - 3)$ to slope-intercept form.

17. A streaming service charges a \$8 monthly fee plus \$3 for each movie rented. Write a linear function for the total cost after x movies. Then find the total cost after 6 movies.
18. A tank contains 500 liters of water and drains at 20 liters per minute. Write a linear function for the amount of water remaining after x minutes. Then find the amount remaining after 12 minutes.
19. A plant is 120 centimeters tall and grows 4 centimeters each month. Write a linear function for the height after x months. Then find the height after 9 months.
20. A delivery service charges a \$6 starting fee plus \$1.75 per mile. Write a linear function for the total cost after x miles. Then find the total cost after 8 miles.